higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

ENGINEERING FABRICATION – SHEET METAL WORK
NQF LEVEL 4

22 NOVEMBER 2019

This marking guideline consists of 6 pages.

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QUESTION 1

1.1 True
1.2 False
1.3 True
1.4 False
1.5 True

(5 × 1) **[5]**

QUESTION 2

2.1 E
2.2 H
2.3 F
2.4 J
2.5 I
2.6 A
2.7 C
2.8 B
2.9 D
2.10 G

(10 × 1) **[10]**

QUESTION 3

3.1
- Spiral duct:
  Is available from 75 mm to 1 500 mm✓✓
- Flex duct:
  Is available from 100 mm to 400 mm✓✓

(2 × 2) (4)

3.2 Thermal acquisition is the control of temperature. (2)

3.3
- Accurate information
- Duct connection
- Service access
- Pipework and electrics
- Airflow access

(5 × 1) (5)

3.4 3.4.1 Forced ventilation is the use of fans and vents to mechanically distribute ventilation and other conditioned air through a building.

3.4.2 Infiltration is the uncontrolled flow of outdoor air into a building.

3.4.3 Natural ventilation is the movement of air into and out of a building through intentional or planned routes without the assistance of mechanical equipment.

(3 × 2) (6)
[17]

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QUESTION 4

4.1 4.1.1 Condensate draining is the process where excess water created during the cooling process is drained away.

4.1.2 The mixing chamber is an area in an air movement system where conditioned air is mixed with outside air.

(2 × 2) (4)

4.2 4.2.1 The flat side of the elbow.

4.2.2 The side that forms the outer curve of the elbow.

4.2.3 The radius of the arc that forms the inner curve/throat of the elbow.

(3 × 2) (6)

4.3
- Check that the correct gas has been chosen.
- Check that the correct polarity has been set.
- Check that you have the correct materials to do the welding.
- Check the power supply.
- Check the wire feed and the current tap tips and look out for any fire hazards. (5)

4.4 0,3 mm✓ or 0,5 mm✓ or 0,7 mm✓ (Any ONE) (1)

[16]

QUESTION 5

5.1 5.1.1
- Argon
- Helium
- Hydrogen (3)

5.1.2 Helium (1)

5.1.3 Red (1)

5.2 5.2.1 Leftward welding (1)

5.2.2 Heat resistance (2)

5.2.3 Tungsten electrodes are non-consumable, because they do not melt and are not transferred to the weld. (2)

5.3 It appears when the hydrogen gas mixes with air and spreads over the highly stressed, hardened part of the weld, resulting in a crack close to the area of the cooler part of the weld. (2)

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5.4
- TIG welding requires a higher level of skill from the welder than other welding processes.
- There is a lower rate of deposition and productivity in TIG welding compared to other welding processes.
- TIG welding equipment is more complex and expensive than other productive processes.
- TIG welding has a higher tolerance for contaminants in filler or base metals.
- Air movement in the area where TIG welding is done may create problems. (5)

[17]

QUESTION 6

6.1
- X axis – Thumb
- Y axis – Fore finger
- Z axis – Middle finger (Axis and indicator to be identified) (3 × 1) (3)

6.2
- On start-up of the machine.
- During power interruptions.
- When the turret accidentally hits the workpiece or a machine component. (3)

6.3
- Chuck
- Cutting tool
- Tailstock
- Bed (4)

6.4 Stock is no longer made ahead of time and stored, but components are fabricated when they are needed. (3)

6.5 6.5.1 Engineering and machining industries:
- In these industries a variety of machines are used to cut materials, shaping two or three-dimensional components.
- Example: CNC grinders do grinding of internal and external components.

6.5.2 Fabrication industries:
- CNC machines are used for cutting and shaping of plates and beams.
- Example: Plasma cutters cutting shapes from plates.

6.5.3 Woodworking industries:
- These industries use CNC machines to cut and shape materials needed in the woodworking industry.
- Example: A common machine used is a wood lathe, shaping objects.

(Any applicable answers acceptable) (3 × 2) (6)

6.6
- Follow all safety procedures regarding the cutting tool.
- Fit the tool and install the workpiece.
- Load the computer programs and key in the required activities on the control panel of the CNC machine. (3)

6.7
- USB Flash disc
- DVD
- Micro disc
- CD (Any 3 × 1) (3)

6.8 6.8.1 Remove all waste material regularly from the bin, to prevent accumulation of swarf.

6.8.2 Separate recyclable and non-recyclable waste.

6.8.3 Follow the company procedures for disposing of non-recyclable and recyclable waste.

(3 × 2) (6)

6.9
- Alignment to bed coordinates.
- Testing for productivity, surface finish and accuracy. (2 × 2) (4)

[35]

TOTAL: 100

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